

Critical success factors for the implementation of urban regeneration REITs in China: a TISM–MICMAC based approach

CSFs for urban
regeneration
REITs in China

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Abstract

Purpose – Real estate investment trusts (REITs) have shown great potential in addressing the current contradiction between underinvestment and sustainable development of urban regeneration in China, as well as in further facilitating the transformation and upgrading of China's urban development. In this regard, this study aims to investigate critical success factors (CSFs) and explore the relationships among these factors, and serve as a reference to provide recommendations and strategies for the successful implementation and sustainable development of urban regeneration REITs.

Design/methodology/approach – In this study, an integrated total interpretive structural modeling–matriced impact croises multiplication applique (TISM–MICMAC) approach using the TISM technique and MICMAC analysis is then implemented to explore the relationships among CSFs, demonstrate the hierarchical structure and classify these factors into clusters based on calculated driving powers and dependence.

Findings – This study has determined a final list of 11 CSFs through literature review and expert survey. The TISM model demonstrates a six-level hierarchical structure encompassing the influence transmission paths of CSFs, in which the most significant factors and links are established, while the MICMAC analysis further classifies CSFs into four clusters as a complement for the findings of the TISM technique.

Practical implications – This study offers practical implications for governments, individual and institutional investors, REITs and property managers, and other stakeholders concluded in urban regeneration REITs. The final list of determined CSFs can serve as the decision points for management and control of the implementation processes, while the findings of the TISM–MICMAC approach can be a significant reference to provide strategies for optimization and enhancement of urban regeneration REITs.

Originality/value – This study is a novel attempt to use both the TISM technique and MICMAC analysis to investigate CSFs for the implementation of urban regeneration REITs, and to address the theoretical and methodological research gaps in the existing literature.

Keywords Sustainable urban development, Urban regeneration REITs, Critical success factors (CSFs), Total interpretive structural modeling (TISM), MICMAC analysis

Paper type Research paper

1. Introduction

Along with the rapid urban development of the past decades, cities have also experienced severe problems such as urban degeneration (Huang *et al.*, 2020), which are required be addressed to ensure sustainable urban development (Abass and Kucukmehmetoglu, 2021).

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The term “urban regeneration” refers to the improvement of the physical, socio-economic and environmental dimensions of cities through various means, including demolition and redevelopment, comprehensive improvement and functional changes (Zheng *et al.*, 2014), in response to the decay process of old urban areas (e.g. old residential and industrial areas) (Zhou *et al.*, 2017a). Therefore, urban regeneration can serve as a fundamental mechanism to effectively utilize urban land resources, improve urban functions, enhance urban image and achieve sustainable urban development (Turcu, 2012).

According to the data from the NBSC (National Bureau of Statistics of China), China’s urbanization rate has reached 64.72% in 2021, while the urban population has exceeded 900 million (NBSC, 2022), indicating that China has formed a huge urban volume and the majority of this country’s population is associated with cities. In recent years, as China’s urbanization process has moved from the “incremental expansion stage” to the “stock renewal stage”, urban regeneration can accordingly be a feasible path to facilitate urban development in China. And over 80% of the urban villages and other decaying areas are currently distributed in China’s relatively developed cities (Pan and Du, 2021), forming a remarkable demand for urban regeneration in these areas. Thus, China’s urban regeneration has been extensively promoted and developed in numerous relatively developed cities, including Shenzhen (Yi *et al.*, 2017), Chongqing (Zhuang *et al.*, 2019), Nanjing (Zhang *et al.*, 2021) and Guangzhou (Lin and De Meulder, 2012).

However, underinvestment frequently leads to stagnant urban regeneration projects (Olanrewaju, 2001), and is widely accepted as the major impediment to the practices and development of urban regeneration. Meanwhile, the large investment requirements of urban regeneration have placed a serious burden on government finances (Pritchett, 2003). For instance, in Shenzhen, influenced by a variety of reasons including underinvestment, only 174 of 509 urban regeneration projects listed by local government were executed from 2010 to 2015 (Liu *et al.*, 2020). Therefore, promoting the marketization of urban regeneration is necessary to address the current contradiction between insufficient investment and sustainable development of urban regeneration (Karaman, 2013), and achieve the urban transformation and upgrading in China.

Real estate investment trusts (REITs) are a type of financial instrument, regarding the properties that can generate stable cash flow and with value-added potential as underlying assets. REITs raise capital by issuing certificates of beneficial interest in the market, entrusting professional institutions to manage the operation of properties and distributing income to investors in proportion (Allen and Sirmans, 1987). Moreover, as an investment products issued and transacted in markets, REITs possess characteristics of low risk, high liquidity and high transparency (Jain and Upadhyay, 2021), and are widely accepted by small- and medium-sized investors (Nichols and Boutell, 2005). From another perspective, REITs are a financing tool that can effectively revitalize the stock of assets and realize the exit of original capital through the real sale of assets, thus reducing the investment risk burden of investors. Therefore, urban regeneration REITs have shown great potential in attracting participation of investment from both individual and institutional investors, facilitating the marketization of urban regeneration and its sustainable development (Allen and Sirmans, 1987; Jain and Upadhyay, 2021). As a result, REITs have been adopted in some large urban regeneration projects in France (Wijburg, 2019), while some scholars suggest to promote the application of REITs to urban regeneration projects in Turkey (Erol and Ozuturk, 2011) and Korea (Song *et al.*, 2021). At the same time, China has preliminarily explored several projects of urban regeneration REITs, such as the issued “Gohighfund urban regeneration equity housing trust” and “GLP China Office Value Add Fund I”.

Although urban regeneration REITs have been promoted at the practical level, there are still deficiencies at theoretical level that make it difficult to guide the long-term sustainable development. A brief review of the prior literature has revealed theoretical and

methodological limitations of existing studies. Firstly, the number of published research articles on urban regeneration REITs is insufficient, while the research content is limited, mainly focusing on the design of the operational model (Deng and He, 2016; Kwon and Bang, 2021) and theoretical framework (Erol and Ozuturk, 2011). Secondly, in extant studies, there is the absence of systematic and thorough research using multicriteria decision-making (MCDM) methods, e.g. the analytic hierarchy process (AHP) and technique for order preference by similarity to ideal solution (TOPSIS) (Sarkar, 2011), making it difficult to provide referable recommendations and strategies for the successful implementation and sustainable development of urban regeneration REITs. Therefore, an urgent demand for an in-depth and thorough analysis of impact mechanism and influence transmission paths has risen to fill this knowledge gap.

This study aims to address the aforementioned theoretical and methodological research gaps by investigating critical success factors (CSFs) for the implementation of urban regeneration REITs in China. To achieve the research goals, the proven effective and feasible techniques of total interpretive structural modeling (TISM) and matrixed impact croises multiplication applique (MICMAC) are employed in this study, which have been widely adopted in various management fields (Gardas *et al.*, 2019; Azadnia *et al.*, 2021; Foli, 2022; Poonia *et al.*, 2022). Specifically, using TISM–MICMAC based approach, this study is a novel attempt to explore the inter- and multi-relationship among CSFs, and demonstrate the hierarchical structure and influence transmission paths of the system. This study would contribute to the current literature by investigating CSFs, and providing references to develop management strategies for optimization and enhancement of urban regeneration REITs. The successful implementation of urban regeneration REITs can effectively address the current challenges faced by urban regeneration in China, and facilitate the transformation and upgrading of China's urban development. More importantly, this study provides inspiration for urban regeneration in other developing countries around the world, and urban regeneration REITs can be a referenceable and effective option to promote their urban development and upgrading.

The rest of this article is organized as follows: identification of CSFs through literature review is conducted in Section 2, while Section 3 describes the research methodology in detail. In Section 4, the processes TISM technique and MICMAC analysis are demonstrated, whereas findings and implications are described in Section 5 and Section 6, respectively. Ultimately, Section 7 depicts the conclusion of this study.

2. Literature review

Although urban regeneration REITs have been explored and developed at both theoretical and practical level, it is still a comparative nascent area of research in general. The current exploration is still at the preliminary stage, mainly focusing on the design of the operational model and theoretical framework, and there is an urgent need for a systematic analysis of impact mechanism for urban regeneration REITs.

CSFs are factors that can impose significant influences on the success of projects (Alias *et al.*, 2014), can support policy makers in developing project implementation strategies (Jenster, 1987) and are widely adopted in various fields such as urban renewal and construction management (Alias *et al.*, 2014; Zhou *et al.*, 2017a). Therefore, this study attempts to identify and understand CSFs for urban regeneration REITs, and accordingly propose effective management measures to guide their successful implementation and development.

The 11 CSFs (summarized in Table 1) identified from the literature are divided into five groups based on the topics covered, namely, government, relationship, project, environment and REITs management. And these factors are discussed further as follows:

Table 1.
CSFs for urban
regeneration REITs
identified from the
literature

Groups	Factors	Literature support
Government	F1: powerful governmental support	Zhang (2005), Newell and Osmadi (2010), Yu and Kwon (2011), Gudiené <i>et al.</i> (2014), Debela (2019) and Zhuang <i>et al.</i> (2019)
Relationship	F2: appropriate supervision and guidance	Walker and Smith (1995), Li (2003), Chan <i>et al.</i> (2010) and Debela (2019)
	F3: effective cooperation mechanism	Larson (1995), Zhang (2005), Oh and Kim (2007), Yu and Kwon (2011), Zheng <i>et al.</i> (2014), Liu <i>et al.</i> (2015) and Chen <i>et al.</i> (2022)
	F4: adequate coordination efforts	Mohsini and Davidson (1992), Merna and Dubey (1998), Zhang (2005), Yu <i>et al.</i> (2006), Yu and Kwon (2011) and Liu <i>et al.</i> (2015)
Project	F5: remarkable locational conditions	Chan and Lee (2008), Newell and Osmadi (2010), Gudiené <i>et al.</i> (2014), Zhou <i>et al.</i> (2017a) and Chen <i>et al.</i> (2022)
	F6: outstanding decision-making and design	Lee and Chan (2008), Yu and Kwon (2011) and Di Maddaloni and Davis (2017)
	F7: satisfactory operational effectiveness	Qiao <i>et al.</i> (2001), Zhang (2005), Yuan <i>et al.</i> (2009) and Liu <i>et al.</i> (2015)
Environment	F8: stable economic environment	Chua <i>et al.</i> (1999), Zhang (2005), Newell and Osmadi (2010) and Smales (2016)
	F9: ideal market environment	Newell and Osmadi (2010), Wu <i>et al.</i> (2011) and Soriano Llobera and Roig Hernando (2015)
REITs management	F10: trusted management capacity	Newell and Osmadi (2010), Soriano Llobera and Roig Hernando (2015) and Kim <i>et al.</i> (2019)
	F11: established information disclosure system	Business (1996), Bushman and Smith (2003), Zhang (2005), Liu <i>et al.</i> (2018) and Chen <i>et al.</i> (2022)

2.1 F1: powerful governmental support

The government plays the role of the facilitator and governor for the implementation of urban regeneration REITs, reflected in the formulating encouraging policies, improving the institutional systems, enhancing administrative efficiency and optimizing related procedures. Specifically, the government can provide flexible and specific support policies, including tax reduction and compensation, appropriate extension of the concession period and payment adjustment mechanism (Zhang, 2005), or assume risk guarantees, such as legal changes, delays in various licenses and certain force majeure. In particular, tax incentives for issued REITs can be an important safeguard to increase the yield and facilitate the successful implementation (Newell and Osmadi, 2010). Moreover, the sound institutional systems, including the legal protection system, financial policies, supervision mechanisms and stakeholder participation policies, can clarify the rights and obligations of each participant (Gudiené *et al.*, 2014; Debela, 2019), restrict the behavior of stakeholders, ensure the rational design and construction of projects (Zhuang *et al.*, 2019) and regulate REITs transactions and operations after issuance. Additionally, efficient administrative efficiency can shorten the construction cycle (Yu and Kwon, 2011) and optimize the issuance procedures of urban regeneration REITs. Given that most phases of urban regeneration projects require local government approval, streamlining cumbersome procedures can help to speed up capital turnover to ensure the continued efficient operation of the projects. Thus, powerful governmental support (F1) is considered to be a critical driver for urban regeneration REITs.

2.2 F2: appropriate supervision and guidance

This factor refers to the appropriate supervision and guidance imposed on urban regeneration REITs. The government assumes part of the governance and guidance

responsibility for urban regeneration projects (Li, 2003). Specifically, it can regulate industry-specific laws, guidelines, standard bidding documents and contract models related to REITs (Debela, 2019). By establishing and optimizing an appropriate regulation system, the government can ensure the timely and faultless delivery of underlying assets, facilitating the effective application of REITs in urban regeneration projects (Chan *et al.*, 2010). On the other hand, overregulation should be avoided as it can create a burden and frustration for the adoption of REITs (Walker and Smith, 1995). Flexible techniques are required to be employed to impose appropriate supervision and guidance (F2) on the transaction and operation processes of urban regeneration REITs, thus facilitating their successful implementation.

2.3 F3: effective cooperation mechanism

Cooperation mechanism is significant throughout the life cycle of the urban regeneration projects (Chen *et al.*, 2022). Compared with other construction projects, urban regeneration projects involve more participants and stakeholders, have great complexity and uncertainty and have remarkable demands on the coordination of stakeholders' interests (Yu and Kwon, 2011; Liu *et al.*, 2015). Usually, urban regeneration projects contain a variety of stakeholders, including all levels of government, institutional and individual private sectors, as well as the public (Zheng *et al.*, 2014), who invariably have difficulty agreeing on the scope, development direction, goals and implementation procedures of the project (Oh and Kim, 2007). Moreover, the introduction of REITs relies on the alignment of benefit distribution between property owners and operators, to reach an agreement on operating procedures. Therefore, an effective cooperation mechanism (F3) is essential for the stable management and development of urban regeneration REITs, associated with participants working together as a team, setting common goals, developing cooperative problem-solving procedures and applying appropriate cooperation techniques (Larson, 1995; Zhang, 2005).

2.4 F4: adequate coordination efforts

Building on the established cooperation mechanism for urban regeneration REITs, coordination efforts are still required to minimize conflicts and disputes among stakeholders (Yu *et al.*, 2006; Yu and Kwon, 2011; Liu *et al.*, 2015). Otherwise, conflicts and disputes may exert great negative impacts on the project performance (Mohsini and Davidson, 1992). Coordination efforts are implemented through contractual arrangements, including agreements between the government and the concessionaire, e.g. concession agreements, shareholder agreements, loan agreements and insurance agreements, and agreements between the concessionaire and the relevant stakeholders, e.g. supply agreements, operating agreements and off-take agreements (Merna and Dubey, 1998). And the aforementioned agreements can effectively avoid disputes over property rights and liabilities, realize risk management (Zhang, 2005) and ultimately ensure the successful implementation of urban regeneration REITs. Thus, adequate coordination efforts (F4) can act as a troubleshooter for various stakeholders in urban regeneration REITs.

2.5 F5: remarkable locational conditions

Locational conditions of urban regeneration projects, including material and cultural conditions (Chen *et al.*, 2022), are the basis of REITs issuance and performance (Newell and Osmadi, 2010). Consisting of appropriate demographic, complete infrastructure, prosperous location and development potential of the area, the material conditions provide sufficient resources for the operational performance of projects (Gudienė *et al.*, 2014). The cultural conditions of urban regeneration projects are usually associated with customs, historical heritage and industries based on local sociocultural status (Chan and Lee, 2008). Based on

locational conditions from both material and cultural perspectives, the distinctive design of projects can exert a significant influence on the achievement of social expectations and sustainability of yield (Zhou *et al.*, 2017a), which is of great importance to the successful implementation of urban regeneration REITs. Therefore, remarkable locational conditions (F5) play the role in generation of economic benefits, thus enhancing the attractiveness of urban regeneration REITs to investors.

2.6 F6: outstanding decision-making and design

Considering the various locational conditions and participants of urban regeneration projects, it is necessary for the processes of decision-making and design to meet the needs of all stakeholders and maximize to economic benefits (Di Maddaloni and Davis, 2017). Specifically, the normative decision-making process for urban regeneration projects includes research and analysis of site characteristics, regional economy and social culture (Lee and Chan, 2008), rational construction scale and architectural design, clarification of property rights structure and design of commercial operation (Yu and Kwon, 2011). On this basis, with reference to the operational and risk profile, flexible management techniques are required to ensure successful transaction and operation of urban regeneration REITs. Thus, outstanding decision-making and design (F6) can be a key factor, which can exert a significant impact on the long-term economic and social benefits of urban regeneration REITs.

2.7 F7: satisfactory operational performance

Operational management of urban regeneration projects is an important part of the life cycle (Liu *et al.*, 2015). REITs regard the property that can generate stable cash flow as underlying assets, and the operational performance of urban regeneration projects directly determines the economic benefits, which in turn affect the operation and performance of REITs. The management and maintenance processes of urban regeneration projects are long-term and continuous, including dynamic consideration of the demand for products and services, long-term cooperation of suppliers (Zhang, 2005) and effective facility management (Yuan *et al.*, 2009) to maximize economic benefits of projects, generate stable cash flow (Qiao *et al.*, 2001) and ultimately facilitate the operation of urban regeneration REITs. Satisfactory operational performance (F7) is hence treated as a critical factor for remarkable operational performance of urban regeneration REITs.

2.8 F8: stable economic environment

The macro and micro economic environment can impose significant influence on the issuance and transaction of urban regeneration REITs (Newell and Osmadi, 2010). When the regional economic development is in an upward period, the consumption power and investment demand of local residents will increase accordingly (Zhang, 2005). A stable economic environment can significantly reduce investment risks, including inflation and interest rate changes (Chua *et al.*, 1999), and avoid frequent fluctuations in the REITs market. More importantly, predictable REITs market risk can significantly reduce investor's fears (Smales, 2016), and ensure that REITs are sufficiently attractive to consistently bring in diversified investment from private sectors. Therefore, a stable economic environment (F8) is closely related to urban regeneration REITs effectively performing the function of asset revitalization and capital exit.

2.9 F9: ideal market environment

The market environment has a direct influence on the issuance and circulation of REITs (Soriano Llobera and Roig Hernando, 2015), and facilitates institutional and individual

investors to invest in various real estate securities products. Compared to other alternative investment products, REITs are more attractive to investors due to higher yield returns and relatively lower risk based on appropriate market environment (Wu *et al.*, 2011). Thus, the ideal market environment (F9) is associated with promoting private sectors investment in urban regeneration REITs (Newell and Osmadi, 2010).

2.10 F10: trusted management capacity

Excellent performance of REITs comes from first-class operational management, which is centered on REITs managers with trustworthy management capacity and credit (Newell and Osmadi, 2010; Soriano Llobera and Roig Hernando, 2015). Therefore, it is necessary for REITs managers to actively participate in the operation and risk management of REITs, and construct a sound management system for long-term development. Additionally, in response to uncertainties involved in the management process, REITs managers are required to demonstrate risk management awareness and response ability (Kim *et al.*, 2019). In conclusion, the trusted management capacity (F10) of urban regeneration REITs managers is a fundamental guarantee for long-term stable operation and satisfactory performance.

2.11 F11: established information disclosure system

Lack of transparency is a major impediment to the successful operation of REITs (Business, 1996). Limited access for the individual and institutional investors to REITs operation information may strike a sense of mistrust. Therefore, project and REITs managers need to focus on adequate communication and information sharing (Chen *et al.*, 2022), including adequacy, clarity and diversity of information disclosure, and timely response to stakeholders and investors (Liu *et al.*, 2018). Otherwise, the poor transparency may lead to corruption and seriously undermines interests of all participants (Zhang, 2005). Therefore, an established information disclosure system (F11) is also a guarantee for long-term operation of urban regeneration REITs (Bushman and Smith, 2003), through which information disclosure can gain the continued trust of investors and facilitate the achievement of investment sustainability.

3. Methodology

In existing studies, numerous research techniques have been introduced to explore the relationships among factors and structures in a system (Foli, 2022). In this study, TISM technique is implemented to explore the inter- and multi-relationships among CSFs for urban regeneration REITs (Jena *et al.*, 2017), and develop a hierarchical relationship model for them. As an extension of the traditional interpretive structural modeling (ISM) technique to address the absence of links interpretation (Azadnia *et al.*, 2021), the TISM technique can be employed to identify and further interpret the significant links among CSFs based on expert opinions, and develop a complete hierarchical TISM diagraph (Bhanot *et al.*, 2017). In addition, MICMAC analysis is introduced to be a complementary tool for the TISM technique, and classifies CSFs into four clusters based on calculated driving powers and dependence (Gardas *et al.*, 2019).

To achieve the systematic exploration of CSFs for urban regeneration REITs, this study has developed a three-stage process in reaching the outcomes. Figure 1 illustrates the framework of this study in detail.

3.1 Stage 1: identification of CSFs by literature review

In the first stage, CSFs for urban regeneration REITs are identified from the relevant literature, and the description of identified CSFs is mentioned in the literature review section.

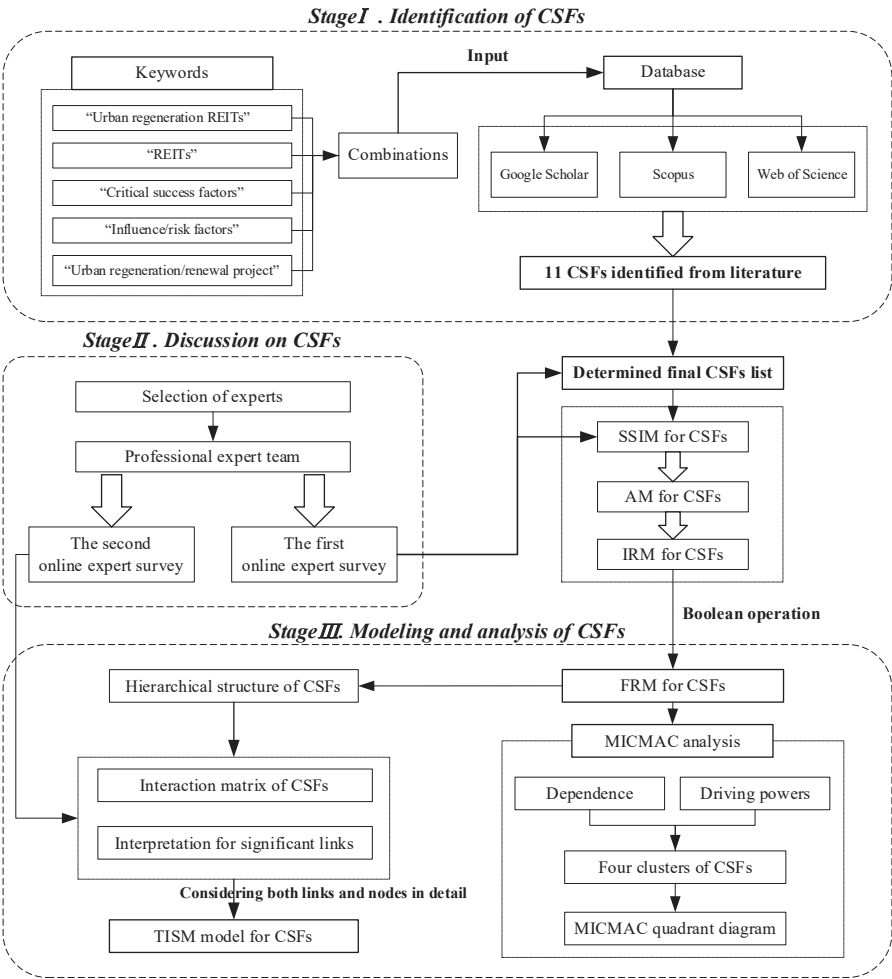


Figure 1.
Framework of
this study

The literature search strategy employed in this study is as follows: 1. Databases frequently used in similar research are selected (Patel *et al.*, 2021; Poonia *et al.*, 2022), including Google Scholar, Scopus and Web of Science; 2. The relevant literature is searched by using the combinations of these keywords, including "Urban regeneration REITs", "Urban regeneration/renewal project", "REITs", "Critical success factors" and "Influence/risk factors."

3.2 Stage 2: discussion on CSFs by expert survey

In the second stage, this study has convened a total of 20 experts to conduct two online surveys. Appendix 1 shows the rationale for the selection of experts and the corresponding information of experts (Table A1). The purpose of the first survey is to examine the rationality of each factor to determine the final list of CSFs, and discuss on the interrelationship among determined CSFs to establish the final reachable matrix (FRM). Meanwhile, by discussing the FRM in depth, the second survey is conducted to develop the

interaction matrix and corresponding interpretation for significant links. The relevant forms of the two expert surveys are attached to the uploaded [Supplementary Materials](#), and the results of the two online surveys are listed in [Appendixes 2 and 3](#), respectively.

3.3 Stage 3: modeling and analysis of CSFs by the TISM technique

In the third stage to modeling and analysis of CSFs, the following stepwise processes are involved:

- (1) Step 1: Based on the final list of CSFs, interrelationships among factors are discussed and confirmed by experts, and the self-interaction matrix (SSIM) is developed according to the following rules for symbols:
 V: if Factor “ i ” influences Factor “ j ”.
 A: if Factor “ j ” influences Factor “ i ”.
 X: if Factors “ i ” and “ j ” influence each other.
 O: if Factors “ i ” and “ j ” are unrelated.
- (2) Step 2: On the basis of the SSIM, the corresponding adjacency matrix (AM) is established according to the following rules:
 - If the cell (i, j) in the SSIM is V, then the cell (i, j) value in the AM is 1 and the cell (j, i) value is 0.
 - If the cell (i, j) in the SSIM is A, then the cell (i, j) value in the AM is 0 and the cell (j, i) value is 1.
 - If the cell (i, j) in the SSIM is X, then the cell (i, j) value in the AM is 1 and the cell (j, i) value is 1.
 - If the cell (i, j) in the SSIM is O, then the cell (i, j) value in the AM is 0 and the cell (j, i) value is 0.
- (3) Step 3: The initial reachable matrix (IRM) is calculated via the AM plus the unit matrix.
- (4) Step 4: The final reachable matrix (FRM) is then calculated by Boolean operations with the support of MATLAB (a programming language developed by MathWorks, Inc.).
- (5) Step 5: Based on the FRM, reachability and antecedent sets of each factor are extracted to establish the hierarchical classification of CSFs. To be specific, the reachability set is the set of F_j corresponding to all cells with value 1 in Row i of the FRM, while the antecedent set is the set of F_i corresponding to all cells with value 1 in Column j of the FRM.
- (6) Step 6: By labeling and interpreting the significant links through the second expert survey, the interaction matrix of CSFs is established.
- (7) Step 7: developing the TISM graphic by considering both links and nodes in detail
- (8) Step 8: Based on the FRM, driving powers and dependence of CSFs are calculated. The driving power is the sum of all “1”s in the corresponding column in the FRM, while the dependence is calculated by summing all “1”s of the corresponding row. And CSFs are classified into four clusters according to the following rules:

Autonomous factors: factors with weak driving power and dependence.

Based on the result of the first online survey, the SSIM (Table 2) is established according to the symbol rules in Section 3. Then the corresponding AM is transformed by SSIM by filling with 1 or 0, depending on the symbols V, A, X and O. After that, the IRM is calculated via the AM plus the unit matrix, and then the FRM is calculated by Boolean operations with the support of MATLAB. Table 3 depicts the FRM of the CSFs, in which direct links among factors are marked as “1” and the transitive links are highlighted by “1*”. Repeatedly extracting reachability and antecedent sets layer by layer, a six-level hierarchical structure (Table 4) is successfully established.

Considering both links and nodes in detail, as well as the hierarchical structure of CSFs, the TISM model is successfully established. As [Figure 2](#) shows that, the direct links are indicated by solid arrows, while significant transitive links are indicated by dashed arrows, and the interpretation statements of corresponding significant links are integrated. In addition, the six levels of factors are filled with different shades of color, with darker colors filling lower-level factors that were located at higher position in the TISM model.

[illegible]

CSFs	1	2	3	4	5	6	7	8	9	10	11	Driving power
F1	1	1	1*	1*	1	1	1	0	1	1*	1	10
F2	0	1	1	1	0	0	1	0	1	1	1	7
F3	0	0	1	1	0	0	1	0	1*	0	1	5
F4	0	0	0	1	0	0	1	0	0	0	0	2
F5	0	0	0	1*	1	1	1	0	0	0	0	4
F6	0	0	0	1	0	1	1	0	0	0	0	3
F7	0	0	0	0	0	0	1	0	0	0	0	1
F8	1	1*	1*	1*	1	1	1	1	1	1*	1*	11
F9	0	0	0	0	0	0	0	0	1	0	0	1
F10	0	1	1	1	0	0	1	0	1*	1	1	7
F11	0	0	0	0	0	0	0	0	1	0	1	2
Dependence power	2	4	5	8	3	4	9	1	7	4	6	53/53

Note(s): *Transitive links

Table 3.
FRM of CSFs

Reachability set	Antecedent set	Intersection	Level
1, 2, 3, 4, 5, 6, 7, 9, 10, 11	1, 8	1	V (1)
2, 3, 4, 7, 9, 10, 11	1, 2, 8, 10	2, 10	IV (2, 5, 10)
3, 4, 7, 9, 11	1, 2, 3, 8, 10	3	III (3, 6)
4, 7	1, 2, 3, 4, 5, 6, 8, 10	4	II (4, 11)
4, 5, 6, 7	1, 5, 8	5	IV (2, 5, 10)
4, 6, 7	1, 5, 6, 8	6	III (3, 6)
7	1, 2, 3, 4, 5, 6, 7, 8, 10	7	I (7, 9)
1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11	8	8	VI (8)
9	1, 2, 3, 8, 9, 10, 11	9	I (7, 9)
2, 3, 4, 7, 9, 10, 11	1, 2, 8, 10	2, 10	IV (2, 5, 10)
9, 11	1, 2, 3, 8, 10, 11	11	II (4, 11)

Table 4.
Hierarchical
classification of CSFs

CSFs	1	2	3	4	5	6	7	8	9	10	11
F1		1 ^d	1 st	1 st	1 ^d					1 st	
F2										1 ^d	
F3				1 ^d							1 ^d
F4							1 ^d				
F5						1 ^d					
F6				1 ^d							
F7											
F8	1 ^d									1 st	
F9											
F10		1 ^d	1 ^d						1 st		
F11									1 ^d		

Note(s): 1^d – direct link
1st – significant transitive link

Table 5.
Interaction matrix
of CSFs

The MICMAC analysis method is then introduced to analyze and interpret CSFs further, as a complementary tool for the TISM technique. Based on the FRM, driving powers and dependence are calculated as listed in Table 3. Then, CSFs are classified into four clusters according to the rules mentioned in Section 3, and the MICMAC analysis quadrant diagram is presented in Figure 3.

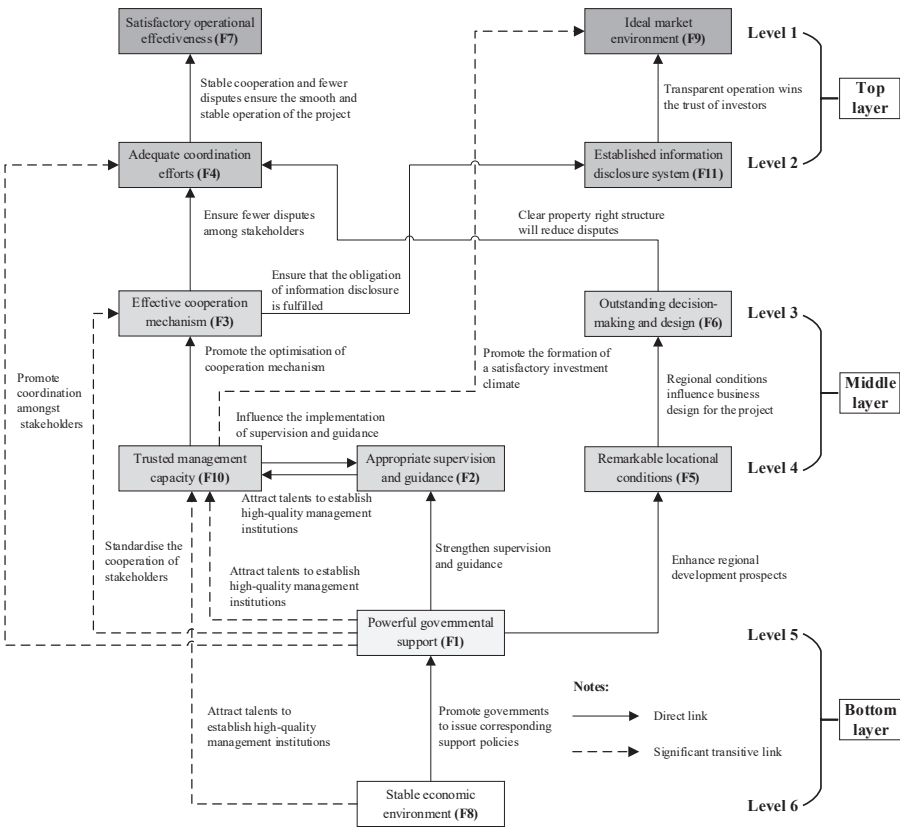


Figure 2.
TISM model of CSFs

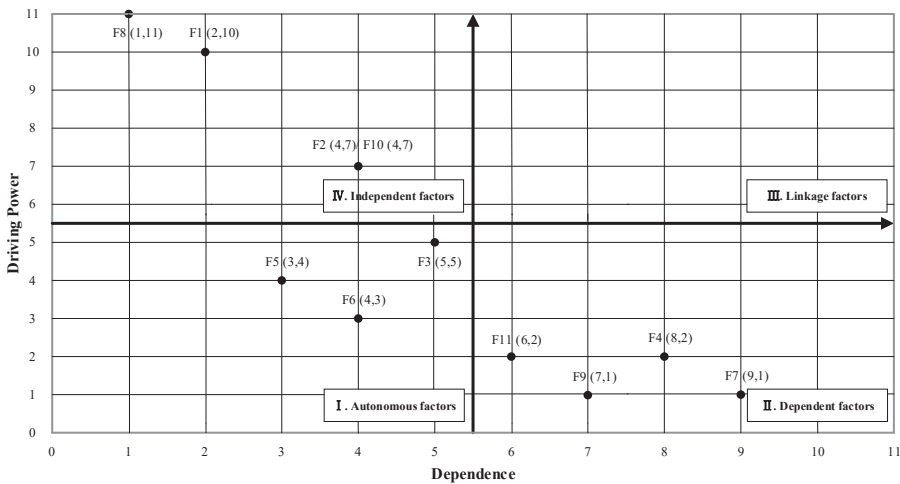


Figure 3.
MICMAC analysis
quadrant diagram

5. Findings

The purpose of this study is to investigate CSFs for urban regeneration REITs, and to further understand the relationships among these factors using TISM–MICMAC based approach. Demonstrating the hierarchical structure and the inter- and multi-relationship among CSFs, the TISM model shown in [Figure 2](#) is strongly associated with the key findings of this study. MICMAC analysis ([Figure 3](#)) is then introduced to classify CSFs into four clusters, i.e. autonomous factors, dependent factors, linkage factors and independent factors, as a complementary tool for the TISM technique. In summary, the findings of this study can be divided into two parts: the hierarchical structure and relationships in TISM model and the classification of CSFs in MICMAC analysis.

5.1 Hierarchical structure and relationships in the TISM model

Based on the influence transmission mechanism demonstrated by significant links, CSFs of the six-level TISM model is divided into three parts as shown in [Figure 2](#), namely, the bottom, top and middle layers, with reference to level and position of each factor. The CSFs located in the bottom layer have the most significant influence on the whole system, including stable economic environment (F8) at Level 6 and powerful governmental support (F1) at Level 5. This part contains factors that have fundamental impacts on the system, and can directly or indirectly influence CSFs in the upper layers. For example, the direct link “F8-F1” indicates that a stable economic environment can drive and facilitate the enhancement of supportive policies, and directly impose influence on governmental support for urban regeneration REITs implementation. This finding of government influence on REITs performance is in response to the study by [Hin Ho et al. \(2013\)](#). In addition, [Ewing and Payne \(2005\)](#) also concluded in their previous study that the economic environment has a fundamental and profound impact on the implementation of REITs, which is directly reflected by remarkable variation of yield and the interest rate.

Occupying the top layer of the CSFs are satisfactory operational effectiveness (F7) and ideal market environment (F9) at Level 1, and adequate coordination efforts (F4) and the established information disclosure system (F11) at Level 2. Based on these factors positioned at the top layer that can directly influence the system performance, direct management strategies can be developed and effectively facilitates the implementation of urban regeneration REITs. For example, the operational effectiveness (F7) of a project is closely associated with the generation of stable cash flows. Considering that the issuance of REITs requires high-quality properties with stable and substantial cash flows as the underlying assets; the operational effectiveness of projects is hence the key factor in the system ([Quek and Eng Ong, 2008](#)). Based on this, corresponding direct management strategies can be developed, such as adopting flexible techniques to optimize the operation process, which can effectively facilitate the successful implementation of urban regeneration REITs. In addition, the top-layer market investment environment (F9) can directly impose influence on the issuance and transaction process of REITs, and attract capital from institutional and individual investors to the long-term operation and development of REITs ([Hardin and Hill, 2008](#)).

The middle layer contains CSFs of Levels 3 and 4, including effective cooperation mechanism (F3) and outstanding decision-making and design (F6) at Level 3, and trusted management capacity (F10), appropriate supervision and guidance (F2) and remarkable locational conditions (F5) at Level 4. Influenced both by bottom-layer factors and influencing top-layer factors, middle-layer factors play the transmission role in the system, which can transmit driving powers and influences from the bottom to the top of the system. For example, material and cultural conditions of the urban regeneration project constitute the locational conditions (F5), which serve as important references for the decision-making and

project design (F6), including architectural design, clarification of property rights structure and design of the commercial operation model (Yu and Kwon, 2011). Meanwhile, clarifying property rights at the decision-making stage helps to reduce possible disputes in the subsequent operation stage (F4), ensures the smooth operation of the project (Liu *et al.*, 2020) and guarantees the successful implementation of urban regeneration REITs.

5.2 Classification of CSFs in MICMAC analysis

MICMAC analysis is incorporated into the TISM technique to classify CSFs into different clusters, i.e. autonomous factors, dependent factors, linkage factors and independent factors. And the position of each factor based on driving power and dependence is demonstrated in Figure 3.

Cluster I consists of “autonomous factors” with weak driving powers and dependence. In this study, three CSFs emerge under this cluster, including remarkable locational conditions (F5), outstanding decision-making and design (F6) and effective cooperation mechanism (F3). Although these factors do not have significant relationship with other factors, they may exert strong influence on the system and require to be considered specially.

Cluster II consists of “dependent factors” with weak driving powers but strong dependence, which are generally located at the top of the TISM model and directly influence the performance of the system. In this cluster, four CSFs are involved, i.e. an established information disclosure system (F11), ideal market environment (F9), adequate coordination efforts (F4) and satisfactory operational effectiveness (F7).

Cluster III consists of “linkage factors” that have strong driving powers and dependence. These CSFs are unstable in the fact that these factors play a role of transmitting influences and receiving feedback (Mohanty, 2018). In this study, no factors emerged under this cluster, and the absence of “Linkage factors” indicates that factors in the system primarily exhibit the single-influence paths. Therefore, the system is relatively easy to implement management and control, on account of avoidance of considering the complexity associated with unstable factors.

Cluster IV consists of “independent factors” with strong driving powers but weak dependence. In this study, four CSFs are obtained under this cluster, including stable economic environment (F8), powerful governmental support (F1), appropriate supervision and guidance (F2) and trusted management capacity (F10). This cluster of CSFs is considered to be the most significant, and some of these factors may even impose direct and indirect influences on the whole system.

6. Implications

6.1 Theoretical implications

This study has addressed the significant theoretical and methodological gaps in the existing literature, and can serve as a reference to provide recommendations and strategies for the implementation of urban regeneration REITs. The critical theoretical implications of this study are summarized as follows:

Firstly, this study systematically investigates the CSFs for the implementation of urban regeneration REITs, and has addressed the research gaps in the existing literature, including the lack of quantity and limitations in research content.

Secondly, based on the TISM–MICMAC based approach, this study is a novel attempt to explore the inter- and multi-relationship among CSFs for urban regeneration REITs, and to demonstrate the hierarchical structure and influences transmission paths of the system. Meanwhile, there are no existing studies on urban regeneration REITs using the TISM technique; this study is therefore the first to employ this proven feasible methodology (Mohanty, 2018; Kumar *et al.*, 2019; Azadnia *et al.*, 2021; Foli, 2022).

Thirdly, through literature review and expert survey, this study has determined a final list of 11 CSFs, and theoretically provided a framework for systematic analysis of CSFs for urban regeneration REITs, which is of significant implications for subsequent studies on relevant topics.

CSFs for urban
regeneration
REITs in China

6.2 Practical implications

Urban regeneration in China has been highly valued and extensively promoted (Zhou *et al.*, 2017b), whereas an unsolved problem has emerged confronted with the contradiction between underinvestment and sustainable development. The proposal of effective management strategies for optimization and enhancement are required to be based on a systematic discussion of impact mechanism, rather than designing models and frameworks from theoretical perspective. The practical implications are described as follows:

Firstly, the determined CSFs are significant for the participants and stakeholders of urban regeneration REITs, who mainly consist of governments, institutional and individual investors, REITs managers and property managers. To be specific, the list of CSFs for urban regeneration REITs can serve as the decision points for management and control of the implementation processes, as well as the awareness guide for policy makers to better understand the characteristics of the system, further giving priority to these key factors while constructing management frameworks and measures.

Secondly, based on the novel TISM–MICMAC approach, relationships among CSFs for urban regeneration REITs are interpreted and illustrated. The findings of this study show that economic environment (F8) and governmental support (F1) are the most significant factors, which have fundamental impact on other CSFs and the whole system. Therefore, it is required to be implemented with the highest priority to manage these factors, instead of attempting to consider and control all CSFs in the system and result in insufficient efficiency. For example, policy makers from financial governmental sectors can dynamically adjust the support policies for urban regeneration REITs according to the economic environment, while REITs managers can accordingly develop flexible strategies to ensure the stable operation and avoid risks including inflation and interest rate changes. In summary, based on a targeted study on these factors with fundamental impact on the system, the system can be optimized and improved by different participants in a synchronized and collaborative manner. In addition, MICMAC analysis is employed to be a complementary tool for the TISM technique and provide references for the instability and interdependence, which is significant for policy makers to better control and manage the whole system.

The implementation and development of urban regeneration REITs is a feasible path that can effectively address the current challenges faced by urban regeneration in China, and facilitate the transformation and upgrading of China's urban development in the long run. More importantly, this study provides inspiration for urban regeneration in other developing countries around the world, and urban regeneration REITs can be a referenceable and effective option to promote their urban development and upgrading.

7. Conclusion

Urban regeneration REITs have shown great potential in addressing the current contradiction between underinvestment and sustainable development of urban regeneration in China, and the successful implementation of urban regeneration REITs has positive influences on the urban regeneration, transformation and sustainable development in other countries. This study is a novel attempt to investigate CSFs for the implementation of urban regeneration REITs in China, and has achieved a systematic and

thorough exploration of CSFs to address the significant gap of an in-depth analysis of the impact mechanisms. Based on the literature review and expert survey, 11 CSFs for urban regeneration REITs are identified and determined eventually. Then, the TISM technique is implemented to explore the inter- and multi-relationships among these factors, and develop a six-level hierarchical relationship model. With reference to the level and position of each factor and the influence transmission mechanism, the CSFs of the six-level TISM model is divided into three parts, i.e. the bottom layer, top layer and middle layer. Based on calculation of driving powers and dependence, MICMAC analysis is employed to be a complementary tool for the TISM technique to classify CSFs into four clusters, i.e. autonomous factors, dependent factors, linkage factors and independent factors. Based on this, management strategies for optimization and enhancement of the system of urban regeneration REITs are specifically discussed.

Although this study has achieved several significant contributions to the implementation and development of urban regeneration REITs, both theoretically and practically, limitations still exist. And this study can be improved in the following aspects. Firstly, the links developed in the TISM model can be further validated using expert surveys in the course of subsequent studies. Secondly, the number of experts invited to participate in the online surveys is limited, and more experts are required to participate in the future study whilst mathematical methods, such as multimean de-entropy evaluation, are needed to reduce the subjectivity of the TISM-MICMAC approach, ensuring the objectivity of the hierarchical structure and relationships among CSFs.

References

- Abass, A.S. and Kucukmehmetoglu, M. (2021), "Transforming slums in Ghana: the urban regeneration approach", *Cities*, Vol. 116, 103284, doi: [10.1016/j.cities.2021.103284](https://doi.org/10.1016/j.cities.2021.103284).
- Alias, Z., Zawawi, E., Yusof, K. and Aris, N. (2014), "Determining critical success factors of project management practice: a conceptual framework", *Procedia-Social and Behavioral Sciences*, Vol. 153, pp. 61-69, doi: [10.1016/j.sbspro.2014.10.041](https://doi.org/10.1016/j.sbspro.2014.10.041).
- Allen, P.R. and Sirmans, C.F. (1987), "An analysis of gains to acquiring firm's shareholders: the special case of REITs", *Journal of Financial Economics*, Vol. 18, pp. 175-184, doi: [10.1016/0304-405X\(87\)90067-5](https://doi.org/10.1016/0304-405X(87)90067-5).
- Azadnia, A.H., Onofrei, G. and Ghadimi, P. (2021), "Electric vehicles lithium-ion batteries reverse logistics implementation barriers analysis: a TISM-MICMAC approach", *Resources, Conservation and Recycling*, Vol. 174, 105751, doi: [10.1016/j.resconrec.2021.105751](https://doi.org/10.1016/j.resconrec.2021.105751).
- Bhanot, N., Rao, P.V. and Deshmukh, S. (2017), "An integrated approach for analysing the enablers and barriers of sustainable manufacturing", *Journal of Cleaner Production*, Vol. 142, pp. 4412-4439, doi: [10.1016/j.jclepro.2016.11.123](https://doi.org/10.1016/j.jclepro.2016.11.123).
- Bushman, R.M. and Smith, A.J. (2003), "Transparency, financial accounting information, and corporate governance", *Financial Accounting Information, and Corporate Governance. Economic Policy Review*, Vol. 9, pp. 65-87.
- Business, A. (1996), "Special report on Asia's infrastructure boom", *Asian Business*, Vol. 9, pp. 60-69.
- Chan, E. and Lee, G.K. (2008), "Critical factors for improving social sustainability of urban renewal projects", *Social Indicators Research*, Vol. 85, pp. 243-256, doi: [10.1007/s11205-007-9089-3](https://doi.org/10.1007/s11205-007-9089-3).
- Chan, A.P., Lam, P.T., Chan, D.W., Cheung, E. and Ke, Y. (2010), "Critical success factors for PPPs in infrastructure developments: chinese perspective", *Journal of Construction Engineering and Management*, Vol. 136, pp. 484-494, doi: [10.1061/\(ASCE\)CO.1943-7862.0000152](https://doi.org/10.1061/(ASCE)CO.1943-7862.0000152).
- Chen, Y., Han, Q., Liu, G., Wu, Y., Li, K. and Hong, J. (2022), "Determining critical success factors of urban renewal projects: multiple integrated approach", *Journal of Urban Planning and Development*, Vol. 148, 04021058, doi: [10.1061/\(ASCE\)UP.1943-5444.0000775](https://doi.org/10.1061/(ASCE)UP.1943-5444.0000775).

- Chua, D.K.H., Kog, Y.-C. and Loh, P.K. (1999), "Critical success factors for different project objectives", *Journal of Construction Engineering and Management*, Vol. 125, pp. 142-150, doi: [10.1061/\(ASCE\)0733-9364\(1999\)125:3\(142\)](https://doi.org/10.1061/(ASCE)0733-9364(1999)125:3(142)).
- Debela, G.Y. (2019), "Critical success factors (CSFs) of public-private partnership (PPP) road projects in Ethiopia", *International Journal of Construction Management*, Vol. 22, pp. 489-500, doi: [10.1080/15623599.2019.1634667](https://doi.org/10.1080/15623599.2019.1634667).
- Deng, H. and He, J. (2016), "Research on the application of PPP model based on REITs in urban regeneration practice", *Proceedings of the 2016 International Conference on Economics, Social Science, Arts, Education and Management Engineering*, pp. 255-266, doi: [10.2991/essaeme-16.2016.54](https://doi.org/10.2991/essaeme-16.2016.54).
- Di Maddaloni, F. and Davis, K. (2017), "The influence of local community stakeholders in megaprojects: rethinking their inclusiveness to improve project performance", *International Journal of Project Management*, Vol. 35, pp. 1537-1556, doi: [10.1016/j.ijproman.2017.08.011](https://doi.org/10.1016/j.ijproman.2017.08.011).
- Erol, T. and Ozuturk, D.D. (2011), "An alternative model of infrastructure financing based on capital markets: infrastructure REITs (InfraREITs) in Turkey", *Journal of Economic Cooperation and Development*, Vol. 32, pp. 65-88.
- Ewing, B.T. and Payne, J.E. (2005), "The response of real estate investment trust returns to macroeconomic shocks", *Journal of Business Research*, Vol. 58, pp. 293-300, doi: [10.1016/S0148-2963\(03\)00147-4](https://doi.org/10.1016/S0148-2963(03)00147-4).
- Foli, S. (2022), "Total interpretive structural modelling (TISM) and MICMAC approach in analysing knowledge risks in ICT-supported collaborative project", *VINE Journal of Information and Knowledge Management Systems*, Vol. 52, pp. 394-410, doi: [10.1108/VJKMS-09-2021-0205](https://doi.org/10.1108/VJKMS-09-2021-0205).
- Gardas, B.B., Raut, R.D. and Narkhede, B. (2019), "Identifying critical success factors to facilitate reusable plastic packaging towards sustainable supply chain management", *Journal of Environmental Management*, Vol. 236, pp. 81-92, doi: [10.1016/j.jenvman.2019.01.113](https://doi.org/10.1016/j.jenvman.2019.01.113).
- Gudienė, N., Banaitis, A., Podvezko, V. and Banaitienė, N. (2014), "Identification and evaluation of the critical success factors for construction projects in Lithuania: AHP approach", *Journal of Civil Engineering and Management*, Vol. 20, pp. 350-359, doi: [10.3846/13923730.2014.914082](https://doi.org/10.3846/13923730.2014.914082).
- Hardin, W. III and Hill, M.D. (2008), "REIT dividend determinants: excess dividends and capital markets", *Real Estate Economics*, Vol. 36, pp. 349-369, doi: [10.1111/j.1540-6229.2008.00216.x](https://doi.org/10.1111/j.1540-6229.2008.00216.x).
- Hin Ho, K., Rengarajan, S. and Han Lum, Y. (2013), "'Green' buildings and real estate investment trust's (REIT) performance", *Journal of Property Investment and Finance*, Vol. 31, pp. 545-574, doi: [10.1108/JPIF-03-2013-0019](https://doi.org/10.1108/JPIF-03-2013-0019).
- Huang, L., Zheng, W., Hong, J., Liu, Y. and Liu, G. (2020), "Paths and strategies for sustainable urban renewal at the neighbourhood level: a framework for decision-making", *Sustainable Cities and Society*, Vol. 55, 102074, doi: [10.1016/j.scs.2020.102074](https://doi.org/10.1016/j.scs.2020.102074).
- Jain, P. and Upadhyay, A. (2021), "Are REITs more resilient than non-REITs? Evidence from natural experiments", *Japan and the World Economy*, Vol. 58, 101069, doi: [10.1016/j.japwor.2021.101069](https://doi.org/10.1016/j.japwor.2021.101069).
- Jena, J., Sidharth, S., Thakur, L.S., Pathak, D.K. and Pandey, V. (2017), "Total interpretive structural modeling (TISM): approach and application", *Journal of Advances in Management Research*, Vol. 14, pp. 102-109, doi: [10.1108/JAMR-10-2016-0087](https://doi.org/10.1108/JAMR-10-2016-0087).
- Jenster, P.V. (1987), "Using critical success factors in planning", *Long Range Planning*, Vol. 20, pp. 102-109, doi: [10.1016/0024-6301\(87\)90160-9](https://doi.org/10.1016/0024-6301(87)90160-9).
- Karaman, O. (2013), "Urban renewal in Istanbul: reconfigured spaces, robotic lives", *International Journal of Urban and Regional Research*, Vol. 37, pp. 715-733, doi: [10.1111/j.1468-2427.2012.01163.x](https://doi.org/10.1111/j.1468-2427.2012.01163.x).
- Kim, S.H., Noh, S. and Lee, S.K. (2019), "Asset-light strategy and real estate risk of lodging C— corps and REITs", *International Journal of Hospitality Management*, Vol. 78, pp. 214-222, doi: [10.1016/j.ijhm.2018.09.004](https://doi.org/10.1016/j.ijhm.2018.09.004).
- Kumar, H., Singh, M.K. and Gupta, M. (2019), "A policy framework for city eligibility analysis: TISM and fuzzy MICMAC-weighted approach to select a city for smart city transformation in India", *Land Use Policy*, Vol. 82, pp. 375-390, doi: [10.1016/j.landusepol.2018.12.025](https://doi.org/10.1016/j.landusepol.2018.12.025).

- Kwon, H. and Bang, D. (2021), "A study on the financing scheme for the sustainability of urban regeneration projects: focusing on the introduction of the regional REITs system", *Journal of Korea Urban Regeneration Association*, Vol. 7, pp. 5-20.
- Larson, E. (1995), "Project partnering: results of study of 280 construction projects", *Journal of Management in Engineering*, Vol. 1111, pp. 30-35, doi: [10.1061/\(ASCE\)0742-597X:2\(30\)](https://doi.org/10.1061/(ASCE)0742-597X:2(30)).
- Lee, G.K. and Chan, E.H. (2008), "Factors affecting urban renewal in high-density city: case study of Hong Kong", *Journal of Urban Planning and Development*, Vol. 134, pp. 140-148, doi: [10.1061/\(ASCE\)0733-9488\(2008\)134:3\(140\)](https://doi.org/10.1061/(ASCE)0733-9488(2008)134:3(140)).
- Li, B. (2003), *Risk Management of Construction Public Private Partnership Projects*, Glasgow Caledonian University, Glasgow.
- Lin, Y. and De Meulder, B. (2012), "A conceptual framework for the strategic urban project approach for the sustainable redevelopment of 'villages in the city' in Guangzhou", *Habitat International*, Vol. 36, pp. 380-387, doi: [10.1016/j.habitatint.2011.12.001](https://doi.org/10.1016/j.habitatint.2011.12.001).
- Liu, J., Love, P.E., Smith, J., Regan, M. and Davis, P.R. (2015), "Life cycle critical success factors for public-private partnership infrastructure projects", *Journal of Management in Engineering*, Vol. 31, 04014073, doi: [10.1061/\(ASCE\)ME.1943-5479.0000307](https://doi.org/10.1061/(ASCE)ME.1943-5479.0000307).
- Liu, B., Wang, X., Xia, N. and Ni, W. (2018), "Critical success factors for the management of public participation in urban renewal projects: perspectives from governments and the public in China", *Journal of Urban Planning and Development*, Vol. 144, 04018026, doi: [10.1061/\(ASCE\)UP.1943-5444.0000467](https://doi.org/10.1061/(ASCE)UP.1943-5444.0000467).
- Liu, G., Wei, L., Gu, J., Zhou, T. and Liu, Y. (2020), "Benefit distribution in urban renewal from the perspectives of efficiency and fairness: a game theoretical model and the government's role in China", *Cities*, Vol. 96, 102422, doi: [10.1016/j.cities.2019.102422](https://doi.org/10.1016/j.cities.2019.102422).
- Merna, A. and Dubey, R. (1998), *Financial Engineering in the Procurement of Projects*, Asia Law & Practice Publishing, Hong Kong.
- Mohanty, M. (2018), "Assessing sustainable supply chain enablers using total interpretive structural modeling approach and fuzzy-MICMAC analysis", *Management of Environmental Quality: An International Journal*, Vol. 29, pp. 216-239, doi: [10.1108/MEQ-03-2017-0027](https://doi.org/10.1108/MEQ-03-2017-0027).
- Mohsini, R. and Davidson, C.H. (1992), "Determinants of performance in the traditional building process", *Construction Management and Economics*, Vol. 10, pp. 343-359, doi: [10.1080/01446199200000030](https://doi.org/10.1080/01446199200000030).
- NBSC (2022), *China Statistics Yearbook 2021*, National Bureau of Statistics, Beijing.
- Newell, G. and Osmadi, A. (2010), "Assessing the importance of factors influencing the future development of REITs in Malaysia", *Pacific Rim Property Research Journal*, Vol. 16, pp. 358-374, doi: [10.1080/14445921.2010.11104309](https://doi.org/10.1080/14445921.2010.11104309).
- Nichols, M. and Boutell, M. (2005), "The introduction of REITs and their impact on the leisure sector", *Journal of Retail and Leisure Property*, Vol. 4, pp. 295-300, doi: [10.1057/palgrave.rlp.5090227](https://doi.org/10.1057/palgrave.rlp.5090227).
- Oh, S.-K. and Kim, J.-H. (2007), "The determinants of timing on project financing in urban renewal project", *Journal of the Korea Planners Association*, Vol. 42, pp. 247-258.
- Olanrewaju, D. (2001), "Urban infrastructure: a critique of urban renewal process in Badia, Lagos", *Habitat International*, Vol. 25, pp. 373-384, doi: [10.1016/S0197-3975\(00\)00042-4](https://doi.org/10.1016/S0197-3975(00)00042-4).
- Pan, W. and Du, J. (2021), "Towards sustainable urban transition: a critical review of strategies and policies of urban village renewal in Shenzhen, China", *Land Use Policy*, Vol. 111, 105744, doi: [10.1016/j.landusepol.2021.105744](https://doi.org/10.1016/j.landusepol.2021.105744).
- Patel, M.N., Pujara, A.A., Kant, R. and Malviya, R.K. (2021), "Assessment of circular economy enablers: hybrid ISM and fuzzy MICMAC approach", *Journal of Cleaner Production*, Vol. 317, 128387, doi: [10.1016/j.jclepro.2021.128387](https://doi.org/10.1016/j.jclepro.2021.128387).
- Poonia, A., Sindhu, S., Arya, V. and Panghal, A. (2022), "Analysis of drivers for anti-food waste behaviour - TISM and MICMAC approach", *Journal of Indian Business Research*, Vol. 14, pp. 186-212, doi: [10.1108/JIBR-02-2021-0069](https://doi.org/10.1108/JIBR-02-2021-0069).

- Pritchett, W.E. (2003), "The public menace of blight: urban renewal and the private uses of eminent domain", *Yale Law & Policy Review*, New Haven, Vol. 21, p. 1.
- Qiao, L., Wang, S.Q., Tiong, R.L. and Chan, T.-S. (2001), "Framework for critical success factors of BOT projects in China", *The Journal of Structured Finance*, Vol. 7, pp. 53-61, doi: [10.3905/jsf.2001.320244](https://doi.org/10.3905/jsf.2001.320244).
- Quek, M.C.H. and Eng Ong, S. (2008), "Securitising China real estate: a tale of two China-centric REITs", *Journal of Property Investment and Finance*, Vol. 26, pp. 247-274, doi: [10.1108/14635780810871623](https://doi.org/10.1108/14635780810871623).
- Sarkar, B. (2011), "Fuzzy decision making and its applications in cotton fibre grading", in Majumdar, A. (Ed.), *Soft Computing in Textile Engineering*, Woodhead Publishing.
- Smales, L.A. (2016), "Risk-on/Risk-off: financial market response to investor fear", *Finance Research Letters*, Vol. 17, pp. 125-134, doi: [10.1016/j.frl.2016.03.010](https://doi.org/10.1016/j.frl.2016.03.010).
- Song, K.-W., Yun, B.-H. and Nam, J. (2021), "Analysis on the change pattern and dynamic characteristics of location centrality in K-REITs' underlying assets using the spatial statistical model", *Journal of Korea Planning Association*, Vol. 56, pp. 67-91, doi: [10.17208/jkpa.2021.02.56.1.67](https://doi.org/10.17208/jkpa.2021.02.56.1.67).
- Soriano Llobera, J.M. and Roig Hernando, J. (2015), "Decision-making factors in the constitution of a REIT", *The 15th International Conference on Accounting and Auditing*, pp. 1-10.
- Turcu, C. (2012), "Local experiences of urban sustainability: researching Housing Market Renewal interventions in three English neighbourhoods", *Progress in Planning*, Vol. 78, pp. 101-150, doi: [10.1016/j.progress.2012.04.002](https://doi.org/10.1016/j.progress.2012.04.002).
- Walker, C.T. and Smith, A.J. (1995), *Privatized Infrastructure: The Build Operate Transfer Approach*, Thomas Telford, London.
- Wijburg, G. (2019), "Reasserting state power by remaking markets? The introduction of real estate investment trusts in France and its implications for state-finance relations in the Greater Paris region", *Geoforum*, Vol. 100, pp. 209-219, doi: [10.1016/j.geoforum.2019.01.012](https://doi.org/10.1016/j.geoforum.2019.01.012).
- Wu, P.-S., Huang, C.-M. and Chiu, C.-L. (2011), "Effects of structural changes on the risk characteristics of REIT returns", *International Review of Economics and Finance*, Vol. 20, pp. 645-653, doi: [10.1016/j.iref.2010.12.002](https://doi.org/10.1016/j.iref.2010.12.002).
- Yi, Z., Liu, G., Lang, W., Shrestha, A. and Martek, I. (2017), "Strategic approaches to sustainable urban renewal in developing countries: a case study of Shenzhen, China", *Sustainability*, Vol. 9, p. 1460, doi: [10.3390/su9081460](https://doi.org/10.3390/su9081460).
- Yu, J.-H. and Kwon, H.-R. (2011), "Critical success factors for urban regeneration projects in Korea", *International Journal of Project Management*, Vol. 29, pp. 889-899, doi: [10.1016/j.ijproman.2010.09.001](https://doi.org/10.1016/j.ijproman.2010.09.001).
- Yu, A.T.W., Shen, Q., Kelly, J. and Hunter, K. (2006), "Investigation of critical success factors in construction project briefing by way of content analysis", *Journal of Construction Engineering and Management*, Vol. 132, pp. 1178-1186, doi: [10.1061/\(ASCE\)0733-9364\(2006\)132:11\(1178\)](https://doi.org/10.1061/(ASCE)0733-9364(2006)132:11(1178)).
- Yuan, J., Zeng, A.Y., Skibniewski, M.J. and Li, Q. (2009), "Selection of performance objectives and key performance indicators in public-private partnership projects to achieve value for money", *Construction Management and Economics*, Vol. 27, pp. 253-270, doi: [10.1080/01446190902748705](https://doi.org/10.1080/01446190902748705).
- Zhang, X. (2005), "Critical success factors for public-private partnerships in infrastructure development", *Journal of Construction Engineering and Management*, Vol. 131, pp. 3-14, doi: [10.1061/\(ASCE\)0733-9364\(2005\)131:1\(3\)](https://doi.org/10.1061/(ASCE)0733-9364(2005)131:1(3)).
- Zhang, W., Zhang, X. and Wu, G. (2021), "The network governance of urban renewal: a comparative analysis of two cities in China", *Land Use Policy*, Vol. 106, 105448, doi: [10.1016/j.landusepol.2021.105448](https://doi.org/10.1016/j.landusepol.2021.105448).
- Zheng, H.W., Shen, G.Q. and Wang, H. (2014), "A review of recent studies on sustainable urban renewal", *Habitat International*, Vol. 41, pp. 272-279, doi: [10.1016/j.habitatint.2013.08.006](https://doi.org/10.1016/j.habitatint.2013.08.006).
- Zhou, T., Zhou, Y. and Liu, G. (2017a), "Comparison of critical success paths for historic district renovation and redevelopment projects in China", *Habitat International*, Vol. 67, pp. 54-68, doi: [10.1016/j.habitatint.2017.06.008](https://doi.org/10.1016/j.habitatint.2017.06.008).

Zhou, T., Zhou, Y. and Liu, G. (2017b), "Key variables for decision-making on urban renewal in China: a case study of Chongqing", *Sustainability*, Vol. 9, p. 370, doi: [10.3390/su9030370](https://doi.org/10.3390/su9030370).

Zhuang, T., Qian, Q.K., Visscher, H.J., Elsinga, M.G. and Wu, W. (2019), "The role of stakeholders and their participation network in decision-making of urban renewal in China: the case of Chongqing", *Cities*, Vol. 92, pp. 47-58, doi: [10.1016/j.cities.2019.03.014](https://doi.org/10.1016/j.cities.2019.03.014).

Appendix 1

Drawing inspiration from a TISM–MICMAC based study by [Azadnia et al. \(2021\)](#), in which the researchers gathered a total of 20 experts from three types of companies to participate in the survey, and discussed the identified barriers for the implementation of electric vehicles lithium-ion batteries reverse logistics. With reference to expert resources of the research team and feasibility of conducting online surveys, this study accordingly convened three types of 20 experts to establish a professional expert team, and [Table A1](#) demonstrates the corresponding information of the selected experts. Conducting the discussion on CSFs for urban regeneration REITs with the expert team, this study has achieved the exploration of inter- and multi-relationship among factors.

Type	Expert	Position	Work experience (years)	Relevance to this study
Field of real estate economics	A1	Research scholar	10	Research mainly focuses on real estate economics, public–private partnership (PPP) and urban regeneration
	A2	Research scholar	6	Research mainly focuses on real estate economics and urban regeneration
	A3	Department manager	18	As the head of the firm’s urban regeneration sector
	A4	Department manager	15	Mainly engaged in real estate valuation
	A5	Property manager	10	Mainly engaged in property operation and management
	A6	Property manager	7	Mainly engaged in property operation and management
Field of finance	B1	Research scholar	15	Research mainly in the field of finance
	B2	Research scholar	12	Research mainly in the field of finance
	B3	Research scholar	7	Research focuses on residential REITs
	B4	Research scholar	4	Research focuses on infrastructure public REITs in China
	B5	REITs manager	6	Have involved in management and operation of REITs
	B6	REITs manager	3	Have involved in management and operation of REITs
	B7	REITs manager	2	Have involved in management and operation of REITs
	B8	Fund manager	7	Have involved in management and operation of funds
	B9	Fund manager	6	Have involved in management and operation of funds
	B10	Fund manager	6	Have involved in management and operation of funds

Table A1.
Profile of experts

(continued)

					CSFs for urban regeneration REITs in China
Type	Expert	Position	Work experience (years)	Relevance to this study	
Government officials	C7	Official	8	Mainly responsible for the management of urban regeneration projects	383
	C8	Official	4	Mainly responsible for the management of PPP projects	
	C9	Official	7	Mainly responsible for the management of the financial market	
	C10	Official	5	Mainly responsible for the management of the financial market	

Table A1.

Appendix 2

Factors	Result of rationality examination	Factors that can be imposed influence on
F1: powerful governmental support	√ (20 of 20 experts agree on)	F2, F5, F6, F7, F9 and F11
F2: appropriate supervision and guidance	√ (19 of 20 experts agree on)	F3, F4, F9, F10 and F11
F3: effective cooperation mechanism	√ (19 of 20 experts agree on)	F4, F7 and F11
F4: adequate coordination efforts	√ (18 of 20 experts agree on)	F7
F5: remarkable locational conditions	√ (20 of 20 experts agree on)	F6 and F7
F6: outstanding decision-making and design	√ (20 of 20 experts agree on)	F1, F4 and F7
F7: satisfactory operational effectiveness	√ (18 of 20 experts agree on)	None of the factors
F8: stable economic environment	√ (20 of 20 experts agree on)	F1, F5, F6, F7 and F9
F9: ideal market environment	√ (20 of 20 experts agree on)	None of the factors
F10: trusted management capacity	√ (20 of 20 experts agree on)	F3, F4, F7 and F11
F11: an established information disclosure system	√ (18 of 20 experts agree on)	F9

Table A2.
Details of the first
online expert survey

Table A3.
Details of the second
online expert survey

Appendix 3

No.	Relation link	Type	Interpretation
1	F1–F2	Direct link	Strengthen supervision and guidance
2	F1–F3	Significant transitive link	Standardize the cooperation of stakeholders
3	F1–F4	Significant transitive link	Promote coordination among stakeholders
4	F1–F5	Direct link	Enhance regional development prospects
5	F1–F10	Significant transitive link	Attract talents to establish high-quality management institutions
6	F2–F10	Direct link	Ensure the qualified ability and standardized behavior of management institutions
7	F3–F4	Direct link	Ensure minimal disputes among stakeholders
8	F3–F11	Direct link	Ensure that the obligation of information disclosure is fulfilled
9	F4–F7	Direct link	Stable cooperation and minimal disputes ensure the smooth and stable operation of the project
10	F5–F6	Direct link	Regional conditions influence business design for the project
11	F6–F4	Direct link	Clear property right structure will minimize disputes
12	F8–F1	Direct link	Promote government issuance of corresponding support policies
13	F8–F10	Significant transitive link	Attract talents to establish high-quality management institutions
14	F10–F2	Direct link	Influence the implementation of supervision and guidance
15	F10–F3	Direct link	Promote the optimization of the cooperation mechanism
16	F10–F9	Significant transitive link	Promote the formation of an acceptable investment climate
17	F11–F9	Direct link	Transparent operation wins the trust of investors

Supplementary materials

Groups	Factors	1.Is this factor rational to be CSF for the system?	2.Which of the other factors judged to be rational can be influenced by this factor?
Government	F1: Powerful governmental support		
	F2: Appropriate supervision and guidance		
Relationship	F3: Effective cooperation mechanism		
	F4: Adequate coordination efforts		
Project	F5: Remarkable locational conditions		
	F6: Outstanding decision-making and design		
	F7: Satisfactory operational effectiveness		
Environment	F8: Stable economic environment		
	F9: Ideal market environment		
REITs management	F10: Trusted management capacity		
	F11: Established information disclosure system		

Table S1.
Form of the first online
expert survey

Type of relation links	Relation links	1.Is this link significant?	2.Interpretation for the significant link.
Direct links	F1–F2	–	
	F1–F5	–	
	F2–F10	–	
	F3–F4	–	
	F3–F11	–	
	F4–F7	–	
	F5–F6	–	
	F6–F4	–	
	F8–F1	–	
	F10–F2	–	
	F10–F3	–	
	F11–F9	–	
Transitive links	F1–F3		
	F1–F4		
	F1–F10		
	F3–F9		
	F5–F4		
	F8–F2		
	F8–F3		
	F8–F4		
	F8–F10		
	F8–F11		
	F10–F9		

Note(s): Direct links are inherently significant, so the corresponding cells are filled with “–”, which meansthatthe expert doesnot need to fill in these cells
The two forms above are previously in Chinese and have been translated in English

Table S2.
Form of the second
online expert survey

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